HEALTHCARE APPOINTMENT ANALYSIS

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Project Name: Healthcare Appointment Analysis

Project Guide: Ms. Rajshri Thete

Project Objective:

The objective of this project was to transform raw healthcare appointment data into meaningful visual insights. I used Power BI to clean, model, and visualize the data through interactive dashboards, allowing users to identify appointment patterns, patient demographics, no-show behavior, and disease-related trends for better healthcare planning and decision-making.

Exploratory Data Analysis:

- Imported the healthcare appointment dataset.
- Checked and corrected data types for all columns.
- Removed duplicate and invalid records.
- Handled missing values and cleaned inconsistent data.
- Created calculated columns such as Month, Day Name, Age Group, and Weekday/Weekend.
- Explored patient demographics (Age, Gender, Scholarship). .
- Analyzed medical conditions such as Hypertension, Diabetes, Alcoholism, and Handicap.
- Identified top neighbourhoods based on appointment count.
- Prepared a clean and structured dataset for building an interactive Power BI and Excel dashboard.

Data Visualization:

- Designed an interactive dashboard to visualize healthcare appointment data.
- Created KPI cards to display Total Appointments, Average Age, No-Show Rate, and Scholarship Patients.
- Used Bar, Line, Donut, and Area charts to analyze appointment trends and patient demographics.
- Added slicers (Gender, Month, Age Group, Scholarship, and SMS Received) for interactive filtering.
- Visualized No-Show patterns, disease prevalence, and neighbourhood-wise appointments.
- Applied a consistent colour theme, icons, and formatting to improve readability and user experience.
- Developed an interactive dashboard that enables stakeholders to gain insights and support data-driven healthcare decisions.

Performance Analysis:

- Analyzed key healthcare KPIs, including Total Appointments, Average Age, No-Show Rate, and Scholarship Patients.
- Evaluated appointment trends across months, weekdays, age groups, and genders.
- Assessed the impact of SMS reminders on patient attendance.
- Compared appointment distribution across different neighbourhoods.
- Analyzed the prevalence of medical conditions such as Hypertension, Diabetes, Alcoholism, and Handicap.
- Used interactive filters and visualizations to identify trends, monitor performance, and support data-driven healthcare decision-making.

Dashboard Development:

- Designed and developed an interactive dashboard for healthcare appointment analysis.
- Created KPI cards to monitor Total Appointments, Average Age, No-Show Rate, and Scholarship Patients.
- Implemented Bar, Line, Donut, and Area charts to visualize appointment trends and patient demographics.
- Added interactive slicers (Gender, Month, Age Group, Scholarship, and SMS Received) for dynamic filtering.
- Applied a consistent layout, colour theme, icons, and formatting to improve readability and user experience.
- Built an interactive dashboard that enables stakeholders to monitor healthcare performance and make data-driven decisions.

Problem Statement:

Healthcare providers often face challenges in managing large volumes of appointment data, identifying patient no-show patterns, and understanding demographic and medical trends. Without a centralized and interactive reporting system, it is difficult to monitor key performance indicators and make informed decisions. This project aims to develop a Power BI dashboard that transforms raw healthcare appointment data into meaningful insights, enabling stakeholders to analyze appointment trends, patient demographics, no-show behavior, and disease prevalence for improved operational efficiency and data-driven decision-making.

Data Description Source

Source

The dataset was retrieved from **Kaggle** and contains historical Healthcare Appointment Analysis.

Data Modelling

- Imported and transformed the healthcare dataset using Power Query.
- Verified data types and created calculated columns (Month, Day Name, Age Group).
- Developed DAX measures for key KPIs such as Total Appointments, Average Age, and No-Show Rate.
- Built an optimized data model to support accurate analysis, interactive filtering, and efficient dashboard performance.

1. Data Model Overview

The data model was designed to organize the healthcare appointment dataset into a clean and structured format for efficient analysis. Data was imported, transformed using Power Query, and validated by assigning appropriate data types. Calculated columns and DAX measures were created to support key metrics such as **Total Appointments, Average Age, No-Show Rate, and Scholarship Patients**. The optimized data model enables accurate calculations, interactive filtering, and improved dashboard performance, ensuring reliable insights for healthcare decision-making.

Entities and Attributes:

Attribute	Description
Patient ID	Unique identifier for each patient.
Appointment ID	Unique identifier for each appointment.
Gender	Gender of the patient.
Age	Age of the patient.
Neighbourhood	Residential area of the patient.
Scholarship	Indicates whether the patient is enrolled in a welfare program.
Hypertension	Indicates whether the patient has hypertension.
Diabetes	Indicates whether the patient has diabetes.
Alcoholism	Indicates whether the patient has alcoholism.
Handicap	Indicates whether the patient has a disability.
SMS Received	Indicates whether an SMS reminder was sent.
Scheduled Day	Date when the appointment was scheduled.
Appointment Day	Date of the scheduled appointment.
No Show	Indicates whether the patient missed the appointment.
Month	Month extracted from the appointment date.
Day Name	Name of the appointment day (e.g., Monday).
Day Number	Numeric representation of the day of the week.
Age Group	Categorizes patients into age groups.
Weekday/Weekend	Indicates whether the appointment was on a weekday or weekend.

This format is suitable for your **project report, viva, and documentation**.

1. Data Relationships:

- Table: Sheet1 (Healthcare Appointment Data)
- Relationship Type: Single-table model (No relationships)
- Primary Data Source: Healthcare Appointment Dataset
- Analysis: Performed using calculated columns, DAX measures, and slicers within the same table.

Approach

1. **Data import**

   Software used

- Microsoft Power BI Desktop

Process

- Import the Healthcare Appointment CSV dataset.
- Verify column names and data types.
- Load the dataset

2. Data Cleaning

- Check for missing values.
- Remove duplicate records.
- Correct data types.
- Standardize team names.
- Prepare the dataset for visualization.

3. Data Analysis and Visualization

- Analyzed Healthcare appointment data using DAX measures and calculated columns to identify key trends and performance metrics.
- Created interactive visualizations, including KPI Cards, Bar Charts, Line Charts, Donut Charts, and Area Charts, to present insights effectively.
- Used slicers for dynamic filtering by Gender, Month, Age Group, Scholarship, and SMS Received.
- Visualized appointment trends, patient demographics, no-show patterns, disease prevalence, and neighbourhood-wise distributions.
- Designed an interactive dashboard to support data-driven healthcare decision-making through clear and meaningful insights.

Project Result:

1. Data Collection and Preparation

The Healthcare appointment dataset was imported from CSV files downloaded from Kaggle.

The dataset was cleaned using Power Query to remove inconsistencies and prepare it for analysis.

2. KPI Dashboard

Displays:

- **Total Appointments –** Displays the total number of appointments scheduled.
- **Average Age –** Shows the average age of patients.
- **No-Show Rate –** Displays the percentage of patients who missed their appointments.
- **Scholarship Patients** – Shows the total number of patients enrolled in the scholarship program.
- **Interactive Filters –** KPI values update dynamically based on slicer selections such as Gender, Month, Age Group, Scholarship, and SMS Received.
- **Real-Time Insights –** Provides a quick overview of healthcare performance to support informed decision-making.

3. **Donut Chart**

Count of AppointmentID by Gender

Result:
Shows the distribution of appointments between male and female patients.

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4. **Area Chart**

Count of AppointmentID by Month and Age Group

Result:
Displays appointment trends across different months and age groups.

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5. **Clustered Bar Chart**

Count of AppointmentID by Neighbourhood

Result:
Identifies the neighbourhoods with the highest number of appointments.

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6. **Clustered Column Chart**

Count of AppointmentID by Month and No Show

Result:
Compares attended and missed appointments across different months.

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7. **Line Chart**

Count of AppointmentID by Day Name and No Show

Result:
Shows no-show trends across the days of the week.

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8. **Clustered Bar Chart**

Count of AppointmentID by Age Group and No Show

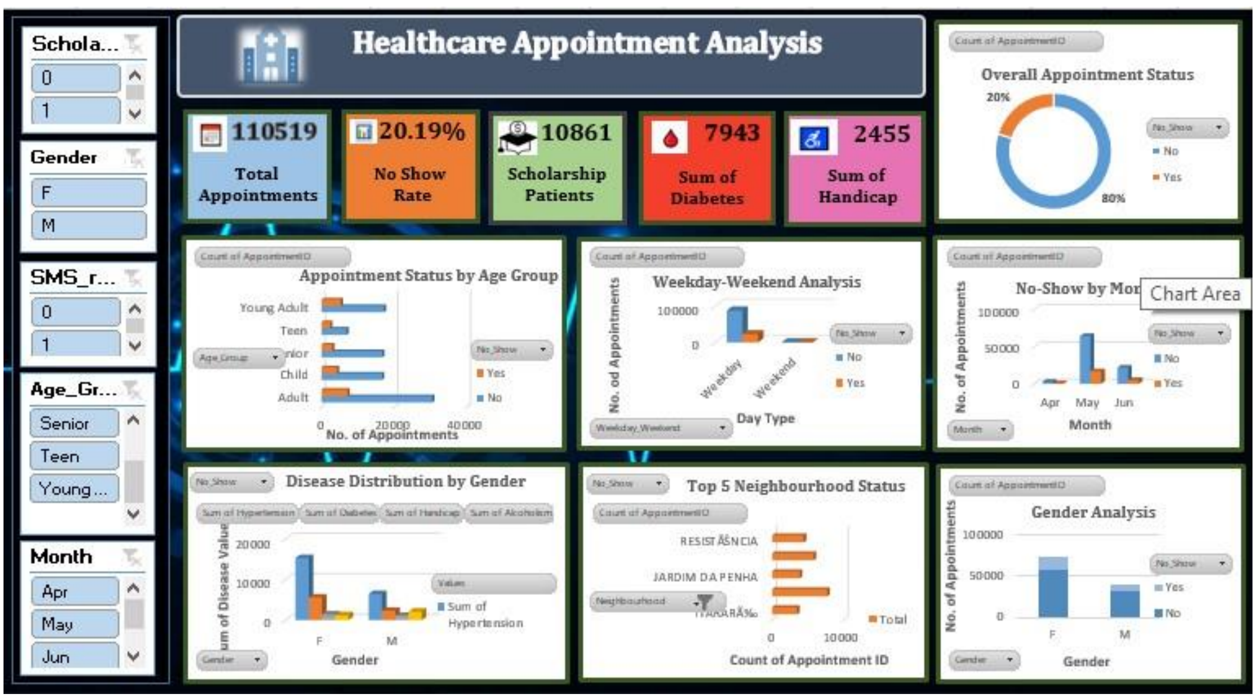
Result:
Compares appointment attendance and no-show patterns across age groups.

9. **KPI Cards**

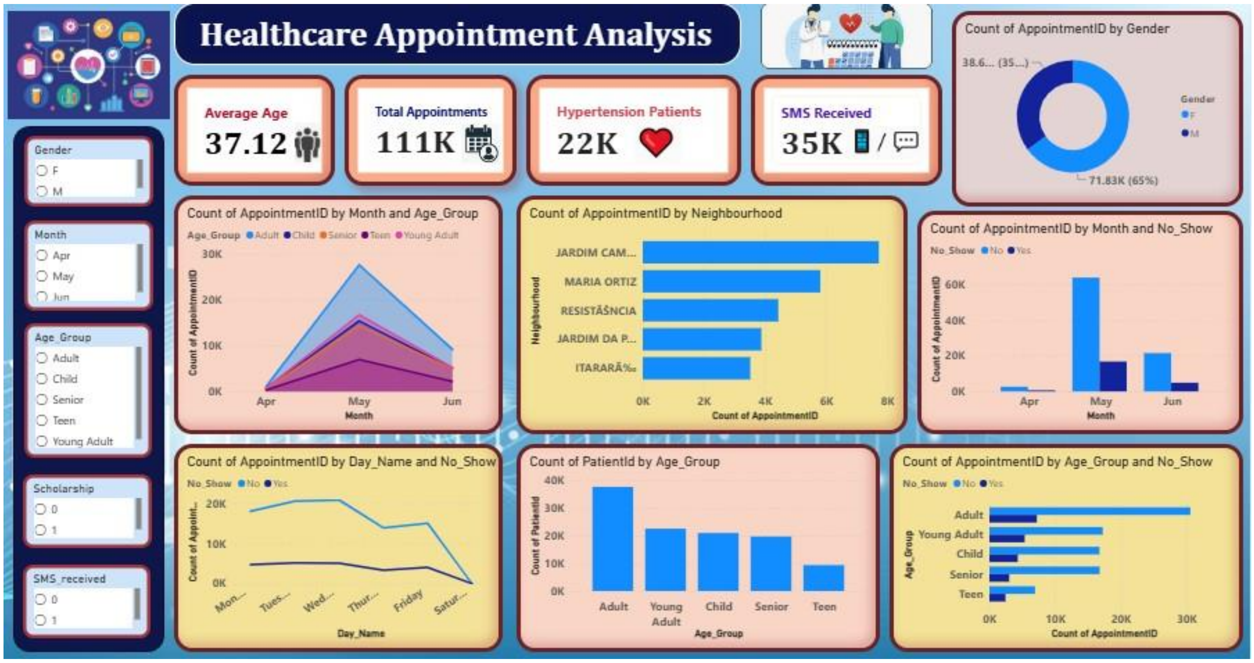
Average Age, Total Appointments, Hypertension Patients, SMS Received

Result:
Provides a quick overview of the key healthcare performance indicators.

Dashboard using Excel:



Dashboard using Power BI



Conclusion:

The **Healthcare Appointment Analysis Dashboard** successfully transforms raw healthcare appointment data into meaningful and interactive visual insights using Power BI. The dashboard enables users to monitor key KPIs, analyze appointment trends, understand patient demographics, evaluate no-show patterns, and assess disease prevalence. Through interactive visualizations and slicers, stakeholders can make informed, data-driven decisions to improve appointment management, optimize healthcare services, and enhance overall operational efficiency.